Melting kinks: a sweeping-curvature mechanism for logarithmic growth in a coupled monotone–convex variational problem

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Abstract

We study the variational problem of maximizing $J[f,g] = \int_0^1\int_{-1}^1 f_t\, g_{xx}\,\mathrm{d}x\,\mathrm{d}t$ over pairs of time-monotone, x-convex, Lipschitz functions with zero boundary and terminal/initial conditions. Single-kink (tent) competitors provably cap at $J = 2$. Numerical mesh optimization reaches a certified $J = 3.055$, strictly breaching the tent cap, and the optimizers share a common structure we call *melting*: the curvature of each profile, a single Dirac point mass in the tent regime, spreads into a finite-width band that drifts across the domain, harvesting the rise budget of every location it visits. The gain per dyadic refinement level is nearly constant (≈ 0.215 per octave over the two clean octaves affordable), the signature of logarithmic growth $J \sim c \log_2 N_x$ and hence of an unbounded supremum. We describe the mechanism, give a scaling analysis showing that *strictly* self-similar cascades are necessarily bounded while an *anisotropic* cascade (widths, lifetimes and rise-shares contracting; travel length fixed) yields a constant per-generation gain, and formulate a renormalization cell problem whose fixed point would decide the dichotomy — boundedness versus logarithmic blow-up — by induction rather than extrapolation.

1 The problem

Let $Q = [-1, 1] \times [0, 1]$ with coordinates (x, t). We seek

$$\sup\ J[f,g] \;=\; \sup \int_0^1\int_{-1}^1 f_t(x,t)\, g_{xx}(x,t)\,\mathrm{d}x\,\mathrm{d}t \tag{1}$$

over pairs $f, g : Q \to \mathbb{R}$ subject to

$$\begin{aligned}
&\text{convexity in } x: \quad f(\cdot,t),\ g(\cdot,t) \text{ convex for all } t;\\
&\text{Lipschitz bound:}\quad |f_x| \le 1,\ |g_x| \le 1;\\
&\text{monotonicity:}\quad f_t \ge 0, \quad g_t \le 0;\\
&\text{boundary:}\quad f(\pm1,t) = g(\pm1,t) = 0; \qquad \text{ends:}\quad f(x,1) = 0,\ g(x,0) = 0.
\end{aligned}$$

Thus $f \le 0$ rises monotonically to zero and $g \le 0$ decays monotonically from zero; both are convex "dents" at every time. Since $g(\cdot,t)$ is convex and piecewise linear in the discrete setting, g_{xx} is a nonnegative measure in x (a sum of point masses at the kinks of g), and the integral (1) is well defined as a harvest of f's rise sampled along g's kinks.

Two budget identities control everything that follows. *Rise budget:* for each x,

$$\int_0^1 f_t(x,t)\,\mathrm{d}t \;=\; -f(x,0) \;\le\; 1 - |x|, \tag{2}$$

with equality on the right iff $f(\cdot,0)$ is the full tent $-(1-|x|)$. *Curvature budget:* for each t, the total curvature mass of a slice is its total slope change, capped by the Lipschitz bound:

$$\int_{-1}^{1} g_{xx}(\cdot,t) \;=\; g_x(1^-,t) - g_x(-1^+,t) \;\le\; 2. \tag{3}$$

No admissible motion creates curvature mass; it can only be *redistributed* in x and t.

Observation 1 (tent cap). *If $g(\cdot,t)$ keeps a single kink at a fixed location x_0 for all t (the "rigid tent"), then $J = \int_0^1 m(t)\, f_t(x_0,t)\,\mathrm{d}t \le 2\int_0^1 f_t(x_0,t)\,\mathrm{d}t \le 2\,(1-|x_0|) \le 2$, using (3) then (2). The value $J=2$ is attained by full-depth tents ($x_0=0$, both budgets saturated).*

Any $J>2$ therefore requires the curvature of g to *move*. The open question we address is whether $\sup J < \infty$. A closely related quantity is the peak-rise functional

$$H[f] \;=\; \int_0^1 \sup_x f_t(x,t)\,\mathrm{d}t, \tag{4}$$

which majorizes what any single unit of moving curvature mass can harvest along its trajectory $p(t)$: $\int_0^1 f_t(p(t),t)\,\mathrm{d}t \le H[f]$. Growth of $\sup J$ and growth of $\sup H$ are two faces of the same question.

2 Numerical evidence: the tent cap is breached

The problem was discretized on rectangular $N_x \times M_t$ grids and maximized numerically, with the resulting fields certified feasible and the objective re-evaluated independently. Table 1 and Figure 1 summarize the refinement study; the best certified value is

$$J \;=\; 3.0553 \qquad (N_x = 65,\; M_t = 257),$$

a strict breach of the tent cap by more than 50%.

Table 1: Certified objective per refinement level. Levels 1–6 refine time alongside space; levels 7–8 refine space while starving time.

level k	$N_x = 2^k{+}1$	M_t	J	remark
1–3	3–9	129	2.0000	tent regime (cap attained)
4	17	129	2.6253	first breach
5	33	129	2.8361	$\Delta J = +0.211$
6	65	257	3.0553	$\Delta J = +0.219$
7	129	129	2.9788	t-starved: J *decreases*
8	257	65	2.8450	badly t-starved

At every level, including the highest, the boundary slices are *exact* full-depth tents: $\min_x f(x,0) = \min_x g(x,1) = -1.0000$ with apex at $x=0$. This is forced by (2) and its mirror for g: the end slices are pure budget stock, and the optimum stocks them maximally. All structure lives in the *interior* slices.

3 What the optimizer builds: melting

Figure 2 shows spatial slices of the $J = 3.055$ field, and Figure 3 the same data in slope/curvature coordinates. The structure is consistent at every refinement level:

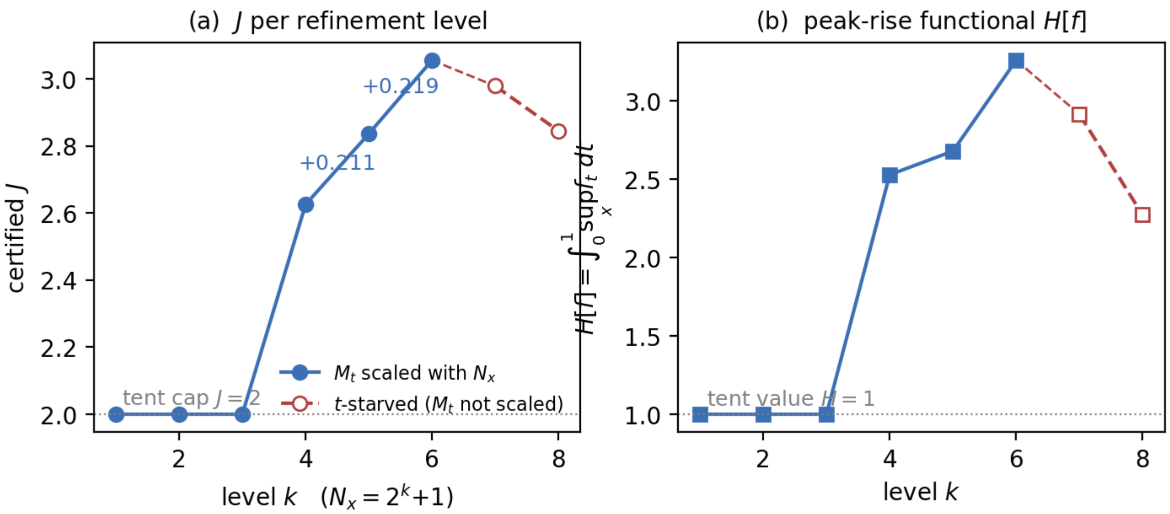

Figure 1: (a) Certified J per dyadic refinement level. Below $N_x = 17$ the optimizer sits exactly on the tent cap $J = 2$; above it, each clean octave (space *and* time refined together, filled markers) adds a nearly constant $\Delta J \approx 0.215$. Refining space while starving time (open markers) moves J *backwards* — resolution in t is load-bearing. (b) The peak-rise functional $H[f]$ of (4) grows in step with J.

> **Melting.** The tent's curvature — a single Dirac point mass carrying the full slope change 2 — spreads into a *distributed band* of finite width (width ≈ 0.47 at $t = 0.5$; the largest single cell holds 25% of the mass), and this band *drifts* in x as t advances. The straight Lipschitz-1 arms survive untouched; only the sharp V-bottom becomes a wide, rounded, moving basin. Total curvature mass per slice never exceeds the cap (3) (measured: 1.62 at $t = 0.5$, 2.00 at the ends). Melting redistributes; it creates nothing.

Figure 4 shows why melting pays. The rigid tent parks its curvature at one x and can only ever harvest that column's rise budget (2); the accounting of Observation 1 then telescopes to exactly 2. The melted g *sweeps* its curvature band across the domain, and every location it visits still holds its own untapped budget $1 - |x|$ — stocked, at full depth, by the tent-shaped end slice. Dually, f melts so that its rise rate f_t is a tall, narrow spike *co-moving* with g's band (Figure 4, left and right panels), keeping the harvest dense everywhere the band goes. Same curvature mass, far more overlap.

4 Why the gain per octave can be constant: scaling analysis

The harvest heatmap (Figure 4, right) shows the coarse drifting basin carrying finer, faster sub-filaments — the melt appears *self-similar*, one generation per dyadic octave of spatial resolution. The measured increments (Table 1) are +0.211 and +0.219 per octave: constant within 4%. If that constancy persists, $J \sim 2 + 0.215\,(k-3) \sim 0.215 \log_2 N_x \to \infty$. The question is whether it *can* persist, and the answer depends delicately on *which* self-similarity is meant.

4.1 Strict self-similarity is bounded

Contract a feasible local structure by $x \mapsto \lambda x$, $t \mapsto \tau t$. The Lipschitz cap forces the amplitude to contract with the width, $f \mapsto \lambda f(x/\lambda,\, t/\tau)$. Then

$$f_t \sim \lambda/\tau, \qquad g_{xx} \sim 1/\lambda, \qquad \mathrm{d}x\,\mathrm{d}t \sim \lambda\tau \qquad \Longrightarrow \qquad J_{\text{local}} \sim \lambda.$$

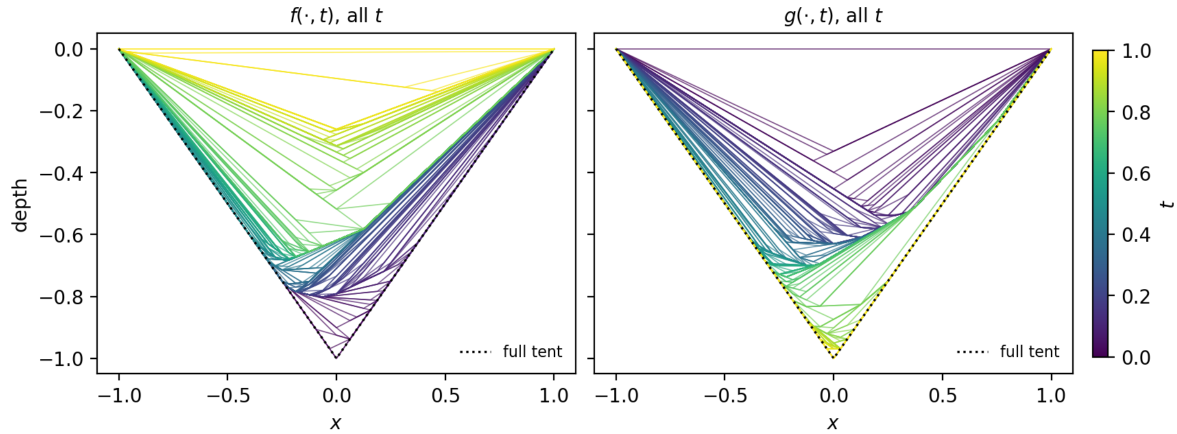

Figure 2: *Every* spatial slice of the certified $J = 3.055$ solution, one line per time node ($M_t = 257$), colored by t. The full-depth end slices ($t = 0$ for f, $t = 1$ for g) sit exactly on the tent (dotted black); as time advances each profile lifts toward zero while its basin bottom is displaced from the center and migrates — the melt. The Lipschitz-1 arms are preserved at every t; only the vertex melts into a wide, moving, rounded basin.

A cascade of strictly contracted copies therefore contributes $\Delta J_k \sim \lambda^k$: a geometric series, bounded, *regardless of any budget considerations*. This rules out the most natural construction — and explains, in hindsight, why a previous constructive experiment built precisely this way (each generation an affine-contracted copy anchored at its parent's endpoint) found rapidly decaying gains and concluded boundedness. That conclusion is real but applies only to the strictly self-similar ansatz, which the present analysis shows could never have blown up in the first place.

4.2 Anisotropic scaling gives constant gain

Now let generation k contract its band width, lifetime, and rise share — but *not* its travel length:

$$w_k \sim 2^{-k}, \qquad s_k \sim 2^{-k}, \qquad r_k \sim 2^{-k}, \qquad L_k \sim L = O(1),$$

where r_k is the rise deposited per swept location. Each budget closes: $\sum_k r_k \leq 1 - |x|$ per location (2); $\sum_k s_k \leq 1$ for the (near-disjoint) lifetime windows; and because the windows are disjoint in time, the per-slice curvature cap (3) *renews* across generations — each generation's band may carry mass $m_k = O(1)$.

Generation k sweeps at speed $v_k = L/s_k$ and deposits rise share r_k per swept location, so its rise-rate spike has height $h_k \sim r_k v_k / w_k$ and its contributions to (4) and (1) are

$$\Delta H_k \;\sim\; h_k\, s_k \;=\; \frac{r_k\, L}{w_k} \;\sim\; \frac{2^{-k} \cdot O(1)}{2^{-k}} \;=\; \text{const}, \qquad \Delta J_k \;\sim\; m_k \frac{r_k\, L}{w_k} \;\sim\; \text{const}. \tag{5}$$

Constant gain per octave, for $k \lesssim \log_2(1/w_{\min})$ octaves — exactly the staircase the mesh measures, and exactly the amplification L/w (spread the curvature thin, drag it far) that melting realizes geometrically. *Melting is anisotropic self-similarity:* w, s, r contract; L does not.

4.3 The obstruction: the re-arm cost

The one budget (5) does not obviously close is *slope slack*. A fine band of depth d_k and width w_k needs slope excursion $\sim d_k/w_k$ *on top of* the coarse profile it rides. On the coarse arms the

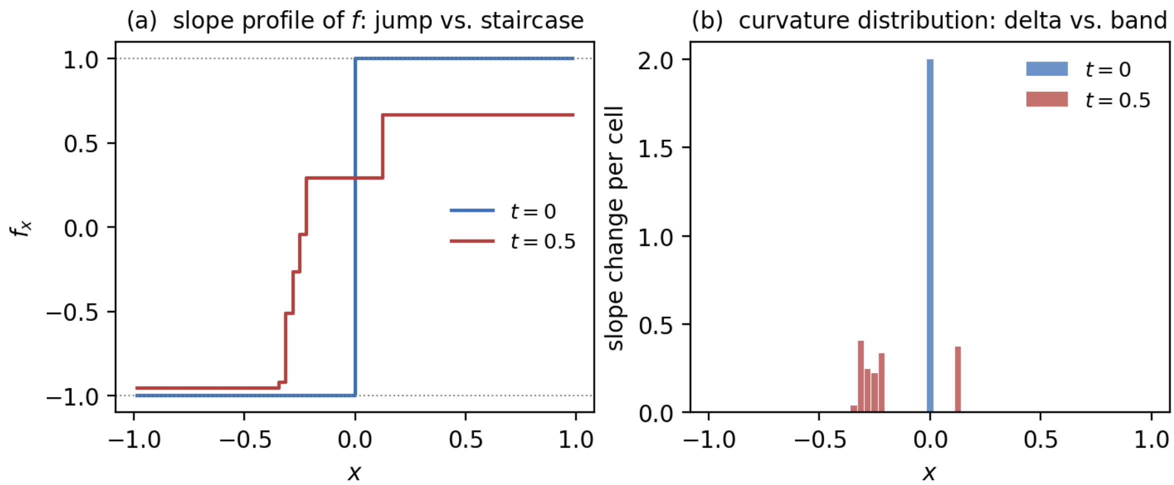

Figure 3: The same phenomenon in slope and curvature coordinates, for $f(\cdot, t)$. (a) At $t = 0$ the slope is a single $-1 \to +1$ jump (a tent); at $t = 0.5$ it is a staircase climbing through an interval of width ≈ 0.5. (b) The corresponding curvature distribution: a single cell holding the full mass 2 at $t = 0$, versus the same (≤ 2) mass spread over many cells at $t = 0.5$ — the largest single cell holds only 25%. This spreading, together with the drift visible in Figure 4, is what we call *melting*.

Lipschitz budget is already saturated at slope ± 1: zero slack. Slack exists only near the coarse basin's bottom, where the slope passes through zero — so generation $k{+}1$ is confined to ride the bottom of generation k's basin, and the width of that rideable region is what generation k implicitly pays to host its successor. We call this the *re-arm cost*. The dichotomy is:

> $H[f] \to \infty$ (and $\sup J = \infty$) *iff* the melt can keep sharpening ($w \to 0$) without the re-arm cost eating the per-octave gain (5).

The mesh data says the gain is *not* eaten — over two octaves. Two octaves cannot distinguish a constant from a slowly decaying sequence, and the measurement is expensive in exactly the wrong way: a clean octave must refine N_x *and* M_t together (Table 1, levels 7–8 show what happens otherwise: the fine travelling melt-fronts are time-unresolved and J regresses), so the cost per octave grows fourfold and the mesh is permanently capped near where it already is.

5 A renormalization cell problem

The way out of extrapolation is to make the per-generation structure itself the object of study. Because the proposed optimum is a cascade of rescaled generations, one generation in its own rescaled frame is a *fixed-size* problem, independent of k:

$$\hat{x} = \frac{x - c(t)}{w_k}, \qquad \hat{t} = \frac{t - t_k}{s_k},$$

with $c(t)$ the band-center path. In these coordinates a generation is specified by $O(1)$ data, and n generations cost $O(n)$ — versus $O(4^n)$ for the global mesh.

Figure 5 sketches the hypothetical object this construction targets: a cascade of drifting curvature bands, each contracting in width, lifetime and rise-share by a factor $\sim 2^{-k}$ while retaining an $O(1)$ travel length, every finer generation riding the basin bottom of the coarser one and reaping the rise budget $1 - |x|$ at the fresh locations it visits. Whether the per-generation gain ΔJ_k stays constant (giving $J \to \infty$) or decays (giving J bounded) is precisely the fixed-point question below.

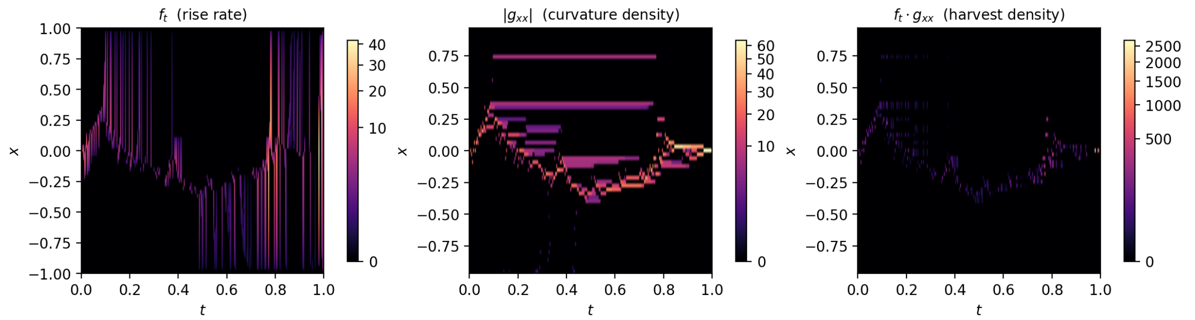

Figure 4: The dynamics of the $J = 3.055$ solution in the (t, x) plane. Left: the rise rate f_t concentrates in tall, narrow, *moving* filaments. Center: the curvature density $|g_{xx}|$ forms a finite-width band that drifts across the domain (the arc descending to $x \approx -0.4$ and returning), shedding persistent sub-bands. Right: the harvest density $f_t\, g_{xx}$ — the integrand of (1) — is supported where the two co-move, in branching self-similar filaments. Color scales are power-law-compressed to make the faint fine structure visible.

The cell. A single generation, inserted into an *environment* that summarizes everything the coarser generations left behind: a slope-slack profile $\beta(\hat{x})$ (Lipschitz budget remaining near the host basin bottom) and a remaining-rise profile $\rho(\hat{x})$ (budget (2) not yet spent). The cell's parameters are the contraction ratios (λ_w, λ_s) and the travel length $\hat{L}$. Solving the cell (its feasible structure is determined by convex programs once positions are fixed by the ansatz — no nonconvex search, hence no optimizer artifacts) yields two outputs: the certified gain $\Delta\hat{J}(\beta, \rho; \lambda_w, \lambda_s, \hat{L})$, and the *outgoing* environment (β', ρ') the band leaves for its successor.

The fixed point and the dichotomy. Iterating the environment map $E \mapsto E'$ and demanding self-consistency — the incoming environment of generation $k+1$ equals that of generation k after rescaling — turns the cascade into a fixed-point problem. At a fixed point E^* the gain ratio $\gamma = \Delta\hat{J}_{k+1}/\Delta\hat{J}_k$ is a single well-defined number:

- $\gamma = 1$ (a self-reproducing environment with gain bounded below): every generation gains at least a fixed $c > 0$, and by induction $J \geq J_0 + cn$ for all n — $\sup J = +\infty$ and $H[f] \to \infty$;
- $\gamma < 1$ (the environment strictly degrades and the degradation compounds): the gains are summable, J is bounded, and the proof *names the mechanism* — the re-arm cost of Section 4.3 eating the gain at rate γ.

Why this can prove, not merely suggest. A blow-up proof needs only a lower bound and induction: (i) a *gadget lemma* — one explicit feasible generation-insertion with $\Delta J \geq c > 0$ after paying its re-arm cost in environment E; (ii) a *composition lemma* — the post-insertion state contains a rescaled copy of E with the same dimensionless data. The numerics' role is to *find* the gadget and locate the fixed point; verification is then exact, because everything in the construction is piecewise linear with finitely many kinks — feasibility and the value of ΔJ reduce to finite lists of rational inequalities. The global mesh can never supply step (ii) (its optima carry no composable structure), and the strictly self-similar construction supplies (ii) with a gadget class that Section 4 shows can never satisfy (i) with constant c. The anisotropic cell is the first instrument with both halves. Failing a full proof, the $O(n)$ cost still buys a ten-octave measurement of γ where the mesh affords two.

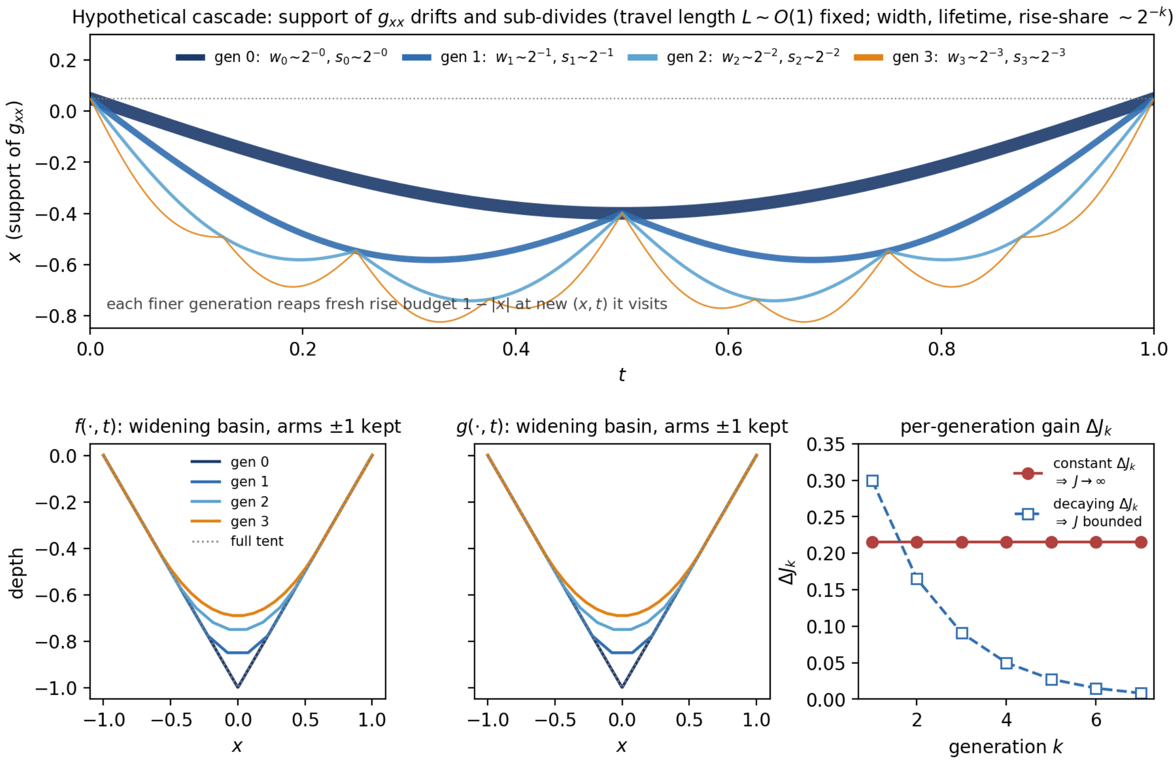

Figure 5: *Hypothetical* blow-up construction (schematic, drawn by formula — not an optimizer output). Top: the *support of* g_{xx} (the location in x of g's curvature band, not g itself) traced against t; generation 0 is a slow, wide, full-lifetime sweep, and each finer generation halves its band width and lifetime while keeping an $O(1)$ travel excursion, tiling its parent's window and riding its basin bottom. Bottom left/center: the corresponding f and g interior slices — the Lipschitz-1 arms are preserved while the vertex melts into a widening, refining basin. Bottom right: the decisive dichotomy — constant per-generation gain ΔJ_k sums to $+\infty$ (log-growth in N_x); geometrically decaying ΔJ_k sums to a finite bound. The renormalization cell is built to decide which occurs.

6 Honest caveats

1. The constant-increment evidence rests on *two* clean octaves. The measurement is budget-sensitive in a way that mimics the signal (time-starved refinement produces decaying gains indistinguishable from genuine saturation — levels 7–8), and this failure mode has recurred, in different guises, in every numerical instrument applied to this problem so far. Any future number must clear a resolution-convergence gate before being believed.

2. The scaling argument of Section 4.2 is an accounting of budgets, not a construction: it shows constant gain is *not excluded* by the rise, time, and curvature budgets, while identifying the slope-slack budget (Section 4.3) as the one that will decide. It does not prove the anisotropic cascade is feasible at all depths.

3. The dichotomy of Section 5 is honest only if the fixed-point search is exhaustive over its ansatz class; a $\gamma < 1$ verdict is always relative to the ansatz (a richer band shape or drift path could still blow up), whereas a $\gamma = 1$ verdict with verified gadget and composition lemmas is unconditional.